

KHUJAND, Tajikistan

WMO: 385990

Lat: 40.215N

Lon: 69.695E

Elev: 427

StdP: 96.30

Time Zone: 5.00 (E05)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-6.8	-12.4	1.4	-5.1	-9.9	1.7	-2.2	12.5	2.0	11.8	1.5	5.3	250	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	38.2	19.1	37.0	18.9	35.9	18.5	20.8	33.7	20.1	32.9	19.5	32.2	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.8	23.7	16.1	12.0	23.5	15.1	11.3	23.7	62.2	34.3	59.5	32.9	57.5	31.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.5	10.5	9.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-10.4	41.1	4.6	1.1	-13.7	41.9	-16.4	42.5	-19.0	43.1	-22.3	43.9
			-10.8	22.4	4.5	0.7	-14.1	23.0	-16.7	23.4	-19.3	23.8	-22.6	24.3
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	1.2	3.2	10.8	16.8	22.7	27.5	29.0	27.2	22.2	15.0	8.0	2.7	(6)
(7)		DBStd	10.48	3.87	4.91	4.36	4.19	3.08	2.65	2.21	2.55	3.04	4.20	4.24	3.50	(7)
(8)		HDD10.0	834	273	192	43	4	0	0	0	0	9	85	228		(8)
(9)		HDD18.3	2186	532	423	233	76	3	0	0	6	118	311	486		(9)
(10)		CDD10.0	2874	0	2	69	208	395	523	588	532	367	164	24	1	(10)
(11)		CDD18.3	1183	0	0	1	28	140	273	330	274	122	14	0	0	(11)
(12)		CDH23.3	14341	0	0	11	285	1379	3431	4443	3408	1254	129	2	0	(12)
(13)		CDH26.7	7514	0	0	1	80	579	1881	2582	1856	511	25	0	0	(13)
(14)	Wind	WSAvg	4.7	5.3	5.5	5.2	4.8	4.4	4.7	4.4	4.2	4.2	4.4	5.2	(14)	
(15)	Precipitation	PrecAvg	162	16	18	26	29	20	5	3	1	2	15	14	17	(15)
(16)		PrecMax	330	58	49	74	85	68	24	22	10	23	90	60	62	(16)
(17)		PrecMin	77	0	2	2	1	1	0	0	0	0	0	0	0	(17)
(18)		PrecStd	57	12	11	18	20	16	6	5	3	5	19	13	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	17.4	24.4	31.8	35.9	40.0	40.8	39.0	35.2	29.2	22.3	13.0	(19)
(20)			MCWB	5.7	9.1	13.0	16.2	17.2	18.7	19.7	19.8	17.4	15.3	13.3	7.7	(20)
(21)		2%	DB	8.9	13.0	21.8	29.0	33.9	37.9	39.0	37.2	33.3	26.8	19.1	10.2	(21)
(22)			MCWB	5.1	7.6	12.3	15.4	17.0	18.4	19.5	19.3	17.0	15.0	11.8	5.8	(22)
(23)		5%	DB	7.1	11.0	19.8	26.8	32.0	36.1	37.2	36.0	31.7	24.2	16.7	8.8	(23)
(24)			MCWB	4.2	6.6	11.6	15.0	16.5	18.0	19.2	18.9	16.7	13.5	10.3	5.1	(24)
(25)		10%	DB	6.0	9.2	17.9	24.2	30.0	34.8	36.0	34.8	29.8	22.2	14.2	7.2	(25)
(26)			MCWB	3.6	5.7	10.9	14.1	16.4	17.7	18.8	18.7	16.0	12.9	9.0	4.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	10.1	14.5	17.7	19.1	20.7	22.0	21.7	19.2	16.5	14.0	8.2	(27)
(28)			MCDWB	8.9	14.4	21.7	25.5	29.8	31.6	35.5	33.4	30.3	25.6	21.6	11.3	(28)
(29)		2%	WB	5.6	8.5	13.4	16.8	18.3	19.8	21.0	20.8	18.4	15.5	12.5	6.7	(29)
(30)			MCDWB	7.7	12.2	19.2	24.5	28.2	31.7	34.4	33.5	29.8	24.1	17.8	8.9	(30)
(31)		5%	WB	4.5	7.2	12.5	16.1	17.7	19.2	20.3	20.0	17.5	14.8	11.1	5.7	(31)
(32)			MCDWB	6.6	10.1	17.9	23.5	27.1	31.3	33.4	32.9	28.9	22.7	15.3	7.7	(32)
(33)		10%	WB	3.5	6.2	11.6	15.2	17.2	18.7	19.7	19.4	16.7	14.0	9.8	4.7	(33)
(34)			MCDWB	5.4	8.6	16.7	22.2	26.3	30.7	32.6	32.0	27.7	20.3	13.5	6.6	(34)
(35)	Mean Daily Temperature Range	MDBR	5.6	6.8	9.6	11.5	12.5	13.5	13.6	14.0	14.1	11.6	8.7	6.0	(35)	
(36)		5% DB	MCDBR	7.0	8.8	12.8	14.9	15.5	15.8	15.6	15.8	16.5	14.2	11.9	7.9	(36)
(37)			MCWBR	4.2	4.8	5.5	5.0	4.2	4.4	4.8	5.4	5.5	5.5	5.5	4.6	(37)
(38)		5% WB	MCDBR	6.5	7.7	11.0	12.0	12.9	13.6	14.2	14.6	15.2	12.5	9.9	7.0	(38)
(39)			MCWBR	4.3	4.7	5.2	4.5	4.0	4.1	4.8	5.2	5.5	4.9	5.1	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.414	0.479	0.498	0.494	0.493	0.479	0.453	0.439	0.477	0.436	0.368	(40)	
(41)		taud	2.153	2.031	1.916	1.920	1.984	1.992	2.032	2.094	2.068	1.913	1.981	2.125	(41)	
(42)		Ebn at Noon	763	773	763	781	795	796	802	812	793	692	661	718	(42)	
(43)		Edh at Noon	106	139	174	185	178	176	168	154	148	155	126	102	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.97	2.76	3.91	5.14	6.40	7.40	7.49	6.83	5.58	3.63	2.23	1.69	(44)	
(45)		RadStd	0.19	0.35	0.36	0.57	0.41	0.31	0.26	0.32	0.19	0.35	0.29	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	+0.51	N/A	N/A	N/A	N/A	+84	+64

Nomenclature: See separate page